A. **Change of Base.** We typically present integers using a base 10 representation.

$\underline{6}\,\underline{3}\,\underline{1}\,\underline{4} = \underline{6} \cdot 10^3 + \underline{3} \cdot 10^2 + \underline{1} \cdot 10 + \underline{4} \cdot 1$

The choice of base 10 is arbitrary, aligning with our fingers or our toes.

The division algorithm, applied sequentially to powers of b, shows

Let b be an integer greater than 1. Then b can be used as a:

This means that every integer n can be uniquely expressed in the form

$n =$

in which case we write

$n =$

where we call $a_m, \ldots, a_2, a_1, a_0$ the **digits** of n **relative base** b.

We can compare sizes of integers by comparing their digits:

So for example, the **binary**, or base 2, representation of 13 is

$23 =$

whereas the **ternary**, or base 3, representation is

$23 =$

All our familar algorithms for computing, by hand, addition, multiplication, and long division, can be converted to work with any base b.

And we can use digits to compare size. For example:

$= (1000)_3 \qquad (222)_3 =$

Some more examples. Base 7 we have

$7^6 =$

And then using the next largest integer base 7 we get

$7^6 - 1 =$

A similar idea base 2 gives

$128 =$

$127 =$

B. **Factorials and Bases.** Say we found the representation of integer n base p.

Then we can give a cleaner expression for the largest power of p that divides $n!$.

Let n be a positive integer and p be:

Write n base p as: $n = (a_m \cdots a_2 a_1 a_0)_p$

Then the exponent of the largest power of p that divides $n!$ is

Proof. From last class, the largest exponent of p that divides $n!$ is given by

Now we establish a lemma.

Lemma. $\left\lfloor \dfrac{(a_m \cdots a_2 a_1 a_0)_p}{p} \right\rfloor =$

Proof. We compute

$$\frac{(a_m \cdots a_2 a_1 a_0)_p}{p}$$

which lets us conclude the greatest integer less or equal $(a_m \cdots a_2 a_1 a_0)_p / p$ is:

□

In the MyOpenMath practice, you proved that

$\left\lfloor \dfrac{x}{m} \right\rfloor =$ whenever m is a positive integer

which we can use to conclude, along with the lemma

$$\left\lfloor \frac{(a_m \cdots a_2 a_1 a_0)_p}{p^2} \right\rfloor =$$

Continuing in this manner, we rewrite our desired exponent

$$\sum_{k=1}^{\infty} \left\lfloor \frac{(a_m \cdots a_2 a_1 a_0)_p}{p^k} \right\rfloor =$$

Example 1. Find the number of zeroes in which $53!$ terminates, as we had done before, but this time working base 5.

Solution. As last time we solved this problem, we need only find the

The base 5 representation of 53 is

$53 =$

and thus the exponent we are looking for is

Example 2. Find the smallest n so $n!$ has 3^{23} as its largest power–of–three factor.

You may express your answer in an alternate base.

Solution. Let's write the base 3 representation of n.

$n =$

Then the maximal power of 3 that divides $n!$ is given by

$23 =$

Since $3^4 - 1 = 80$ we immediately see that:

Thus

$46 =$

By guess–and–check, the smallest that a_2 can be is: $a_3 =$

in which case the smallest that a_2 can be is: $a_2 =$

in which case the smallest that a_1 can be is: $a_1 =$

in which case the smallest that a_0 can be is: $a_0 =$

Thus our final answer is

$n =$

C. **Floor and Convolutions.** The floor function offers rewrites of convolutions.

Theorem. Let f be arithmetic and $F =$

Let n be a positive integer. Then

$$\sum_{k=1}^{n} F(k) =$$

Proof. We examine the sum on the lefthand side.

$$\sum_{k=1}^{n} F(k) =$$

i.e. we are summing $f(d)$ for all

Returning to the general case, let's fix a specific value of d from 1 to n.

The number of times $f(d)$ will occur in our sum is

or in other words equals:

Thus, summing over d first, the sum is equivalent to

$$\sum_{d=1}^{n}$$

□